

FormFix AI

Real-Time Exercise Pose Detection, Rep Counting, and Posture Correction

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The Challenge in Home Fitness

The Supervision Gap

- Improper form increases injury risk.
- Manual rep counting distracts the user.
- Most systems recognize the exercise but do not judge movement quality.

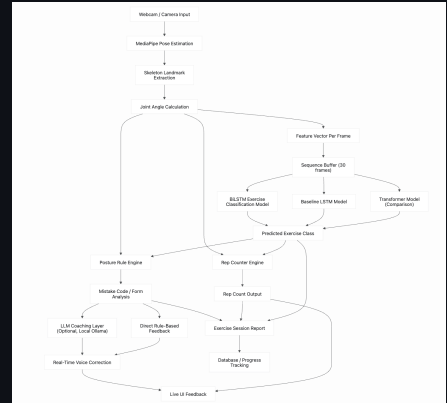
What FormFix AI Solves

- real-time exercise classification,
- accurate rep counting,
- posture correction using joint angles,
- web-based progress tracking and workout reports.

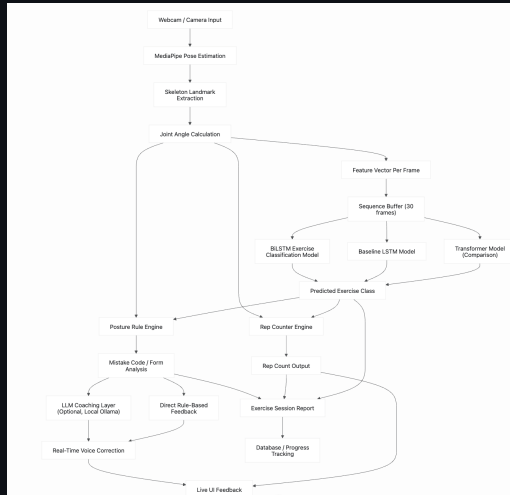
Chapter1: Introduction
Chapter2: Literature Survey: Write about related papers or work in 1 paragraph per paper
Chapter3: Mathematical model/Architecture- System Requirements or Specification
Chapter4: Problem definition: Objective, Assumption
Chapter5: Algorithms and examples
Chapter6: Data structures used/Implementation
Chapter7: Simulation, Results, Performance Analysis, Graphs , Tables
Chapter8: Conclusions (3 Paragraph, 1 page)
Bibliography (IEEE format)
Author, Title, Publisher, page number, year of publication
Appendix

Objectives: Beyond Recognition

- 1 **Real-Time Perception:** detect skeleton landmarks from webcam video.
- 2 **Temporal Intelligence:** compare LSTM, BiLSTM, and Transformer models.
- 3 **Biomechanical Validation:** use joint-angle rules for posture correction.
- 4 **Feedback Loop:** provide live voice and visual guidance.
- 5 **Persistence:** store user workout progress and reports in SQL.



The Hybrid Intelligence Architecture



From Video to Sequential Features

Dataset Specs

- Processed sequences: **1140**
- Features per frame: **132**
- Time window: **30 frames**
- Classes: **6**

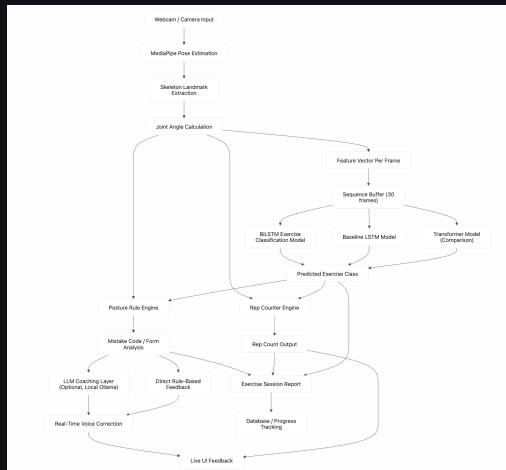
Angle Formula

$$\theta = \cos^{-1} \left(\frac{(A - B) \cdot (C - B)}{\|A - B\| \|C - B\|} \right)$$

Class Distribution

- barbell_squat: 152
- bicep_curl: 225
- hammer_curl: 78

Ahmed & Muradi (UVCE)



Model Strategy: Why Three Models?

Model	Role	Reason for Use
LSTM	Baseline model	Lightweight and fast sequence model for comparison.
BiLSTM	Main temporal model	Uses forward and backward context; stronger motion understanding.
Transformer	Advanced comparison	Uses self-attention for long-range frame relationships.

Deployment logic: the system supports *Fast*, *Best*, *Hybrid*, and *Ensemble* modes. In practice, BiLSTM is weighted highest in the hybrid pipeline.

LSTM, BiLSTM and Transformer Architectures



LSTM: single-direction temporal baseline.



BiLSTM: bidirectional temporal classifier.



Transformer: attention-based sequence model.

Working Principle

Validation Engines: Counting and Correction

Repetition Counting

- exercise-specific tracked angle,
- smoothing and direction gating,
- stage transitions such as up/down,
- threshold validation before incrementing rep count.

Posture Correction

- elbow, knee, shoulder, and hip angles,
- rule templates for each exercise,
- mistake codes such as elbow instability or insufficient depth,
- live voice cue and end-of-exercise report.

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Appendix

Training and Evaluation Snapshots

```
Epoch 1/22
187/187 [=====] - 8s 28ms/step - loss: 1.8406 - accuracy: 0.6286 - val_loss: 0.8800 - val_accuracy: 0.6622 - lr: 0.0010
Epoch 3/22
187/187 [=====] - 4s 28ms/step - loss: 0.8195 - accuracy: 0.7390 - val_loss: 0.7413 - val_accuracy: 0.7130 - lr: 0.0010
Epoch 5/22
187/187 [=====] - 4s 19ms/step - loss: 0.4087 - accuracy: 0.7722 - val_loss: 0.4806 - val_accuracy: 0.8157 - lr: 0.0010
Epoch 6/22
187/187 [=====] - 4s 19ms/step - loss: 0.5918 - accuracy: 0.8184 - val_loss: 0.6236 - val_accuracy: 0.8277 - lr: 0.0010
Epoch 9/22
187/187 [=====] - 4s 28ms/step - loss: 0.5868 - accuracy: 0.8364 - val_loss: 0.5967 - val_accuracy: 0.8485 - lr: 0.0010
Epoch 10/22
187/187 [=====] - 4s 19ms/step - loss: 0.4726 - accuracy: 0.8486 - val_loss: 0.2867 - val_accuracy: 0.8914 - lr: 0.0010
Epoch 7/22
187/187 [=====] - 4s 20ms/step - loss: 0.4130 - accuracy: 0.8667 - val_loss: 0.3230 - val_accuracy: 0.8881 - lr: 0.0010
Epoch 8/22
187/187 [=====] - 4s 20ms/step - loss: 0.3976 - accuracy: 0.8689 - val_loss: 0.2603 - val_accuracy: 0.9076 - lr: 0.0010
Epoch 9/22
187/187 [=====] - 4s 21ms/step - loss: 0.3691 - accuracy: 0.8848 - val_loss: 0.2862 - val_accuracy: 0.8914 - lr: 0.0010
Epoch 10/22
187/187 [=====] - 4s 20ms/step - loss: 0.3684 - accuracy: 0.8984 - val_loss: 0.2482 - val_accuracy: 0.9082 - lr: 0.0010
Epoch 11/22
187/187 [=====] - 4s 20ms/step - loss: 0.3089 - accuracy: 0.9016 - val_loss: 0.2610 - val_accuracy: 0.8920 - lr: 0.0010
Epoch 12/22
187/187 [=====] - 4s 19ms/step - loss: 0.3866 - accuracy: 0.8983 - val_loss: 0.2409 - val_accuracy: 0.9075 - lr: 0.0010
Epoch 13/22
186/187 [=====] - ETA: 0s - loss: 0.3287 - accuracy: 0.8987
Epoch 13: ReduceLROnPlateau reducing learning rate to 0.00050000000000000005
187/187 [=====] - 4s 19ms/step - loss: 0.3282 - accuracy: 0.8989 - val_loss: 0.2587 - val_accuracy: 0.9040 - lr: 0.0010
Epoch 14/22
187/187 [=====] - 4s 20ms/step - loss: 0.2174 - accuracy: 0.9279 - val_loss: 0.1766 - val_accuracy: 0.9236 - lr: 5.0000e-04
Epoch 15/22
187/187 [=====] - 4s 19ms/step - loss: 0.1682 - accuracy: 0.9351 - val_loss: 0.1508 - val_accuracy: 0.9430 - lr: 5.0000e-04
Epoch 16/22
187/187 [=====] - 4s 20ms/step - loss: 0.1877 - accuracy: 0.9403 - val_loss: 0.1379 - val_accuracy: 0.9511 - lr: 5.0000e-04
Epoch 17/22
187/187 [=====] - 4s 19ms/step - loss: 0.1829 - accuracy: 0.9418 - val_loss: 0.1472 - val_accuracy: 0.9424 - lr: 5.0000e-04
Epoch 18/22
187/187 [=====] - 4s 19ms/step - loss: 0.1563 - accuracy: 0.9495 - val_loss: 0.1248 - val_accuracy: 0.9530 - lr: 5.0000e-04
Epoch 19/22
187/187 [=====] - 4s 19ms/step - loss: 0.1512 - accuracy: 0.9506 - val_loss: 0.1172 - val_accuracy: 0.9564 - lr: 5.0000e-04
Epoch 20/22
187/187 [=====] - 4s 20ms/step - loss: 0.1629 - accuracy: 0.9531 - val_loss: 0.1392 - val_accuracy: 0.9491 - lr: 5.0000e-04
Epoch 21/22
187/187 [=====] - 4s 20ms/step - loss: 0.1456 - accuracy: 0.9499 - val_loss: 0.1211 - val_accuracy: 0.9611 - lr: 5.0000e-04
Epoch 22/22
186/187 [=====] - ETA: 0s - loss: 0.1488 - accuracy: 0.9456
Epoch 22: ReduceLROnPlateau reducing learning rate to 0.00025000000000000003
187/187 [=====] - 4s 20ms/step - loss: 0.1677 - accuracy: 0.9454 - val_loss: 0.1199 - val_accuracy: 0.9530 - lr: 5.0000e-04
2026-04-13 00:14:14.317 | train_lstm | INFO | Saved BiLSTM model to models/exercise_bi_lstm.keras
2026-04-13 00:14:14.319 | train_lstm | INFO | Saved LSTM model to models/exercise_lstm.keras
2026-04-13 00:14:14.320 | train_lstm | INFO | Final validation accuracy: 0.9799
```

```
Epoch 1/30
2026-04-13 00:14:44.402117: I tensorflow/core/grappler/optimizers/custom_graph_optimizer_registry.cc:1177 Plugin optimizer for device_type GPU is enabled.
187/187 [=====] - 8s 36ms/step - loss: 0.8476 - accuracy: 0.6743 - val_loss: 0.5449 - val_accuracy: 0.7836
Epoch 2/30
187/187 [=====] - 4s 30ms/step - loss: 0.4489 - accuracy: 0.8364 - val_loss: 0.3719 - val_accuracy: 0.8572
Epoch 3/30
187/187 [=====] - 4s 30ms/step - loss: 0.3516 - accuracy: 0.8699 - val_loss: 0.3181 - val_accuracy: 0.8921
Epoch 4/30
187/187 [=====] - 4s 41ms/step - loss: 0.2845 - accuracy: 0.8964 - val_loss: 0.3335 - val_accuracy: 0.8854
Epoch 5/30
187/187 [=====] - 13s 71ms/step - loss: 0.2680 - accuracy: 0.9084 - val_loss: 0.2444 - val_accuracy: 0.9109
Epoch 6/30
187/187 [=====] - 7s 36ms/step - loss: 0.2148 - accuracy: 0.9179 - val_loss: 0.1676 - val_accuracy: 0.9358
Epoch 7/30
187/187 [=====] - 7s 39ms/step - loss: 0.1892 - accuracy: 0.9298 - val_loss: 0.1714 - val_accuracy: 0.9471
Epoch 8/30
187/187 [=====] - 7s 37ms/step - loss: 0.1999 - accuracy: 0.9211 - val_loss: 0.2646 - val_accuracy: 0.9048
Epoch 9/30
187/187 [=====] - 7s 36ms/step - loss: 0.1762 - accuracy: 0.9316 - val_loss: 0.1764 - val_accuracy: 0.9438
Epoch 10/30
187/187 [=====] - 8s 46ms/step - loss: 0.1610 - accuracy: 0.9273 - val_loss: 0.1876 - val_accuracy: 0.9438
Epoch 11/30
187/187 [=====] - 7s 37ms/step - loss: 0.1608 - accuracy: 0.9383 - val_loss: 0.1861 - val_accuracy: 0.9383
Epoch 12/30
187/187 [=====] - 7s 39ms/step - loss: 0.1458 - accuracy: 0.9468 - val_loss: 0.1584 - val_accuracy: 0.9417
Epoch 13/30
187/187 [=====] - 7s 36ms/step - loss: 0.1187 - accuracy: 0.9641 - val_loss: 0.1449 - val_accuracy: 0.9508
Epoch 14/30
187/187 [=====] - 4s 33ms/step - loss: 0.1087 - accuracy: 0.9588 - val_loss: 0.1647 - val_accuracy: 0.9497
Epoch 15/30
187/187 [=====] - 7s 35ms/step - loss: 0.1328 - accuracy: 0.9469 - val_loss: 0.1492 - val_accuracy: 0.9497
Epoch 16/30
187/187 [=====] - 4s 34ms/step - loss: 0.1117 - accuracy: 0.9568 - val_loss: 0.1562 - val_accuracy: 0.9497
Epoch 17/30
187/187 [=====] - 7s 36ms/step - loss: 0.1064 - accuracy: 0.9688 - val_loss: 0.1378 - val_accuracy: 0.9578
Epoch 18/30
187/187 [=====] - 4s 33ms/step - loss: 0.0923 - accuracy: 0.9645 - val_loss: 0.1246 - val_accuracy: 0.9645
Epoch 19/30
187/187 [=====] - 7s 36ms/step - loss: 0.1082 - accuracy: 0.9581 - val_loss: 0.1489 - val_accuracy: 0.9491
Epoch 20/30
187/187 [=====] - 4s 35ms/step - loss: 0.1186 - accuracy: 0.9584 - val_loss: 0.1598 - val_accuracy: 0.9551
Epoch 21/30
187/187 [=====] - 8s 44ms/step - loss: 0.0912 - accuracy: 0.9661 - val_loss: 0.1786 - val_accuracy: 0.9418
Epoch 22/30
187/187 [=====] - 4s 32ms/step - loss: 0.0908 - accuracy: 0.9661 - val_loss: 0.1256 - val_accuracy: 0.9658
Epoch 23/30
187/187 [=====] - 4s 34ms/step - loss: 0.0817 - accuracy: 0.9698 - val_loss: 0.1816 - val_accuracy: 0.9725
Epoch 24/30
187/187 [=====] - 4s 33ms/step - loss: 0.0781 - accuracy: 0.9732 - val_loss: 0.1421 - val_accuracy: 0.9611
Epoch 25/30
187/187 [=====] - 7s 39ms/step - loss: 0.0691 - accuracy: 0.9779 - val_loss: 0.1351 - val_accuracy: 0.9638
Epoch 26/30
187/187 [=====] - 18s 52ms/step - loss: 0.0909 - accuracy: 0.9626 - val_loss: 0.1683 - val_accuracy: 0.9524
Epoch 27/30
187/187 [=====] - 7s 40ms/step - loss: 0.0898 - accuracy: 0.9676 - val_loss: 0.1596 - val_accuracy: 0.9538
Epoch 28/30
187/187 [=====] - 9s 68ms/step - loss: 0.1029 - accuracy: 0.9614 - val_loss: 0.1360 - val_accuracy: 0.9638
Epoch 29/30
187/187 [=====] - 4s 34ms/step - loss: 0.1128 - accuracy: 0.9589 - val_loss: 0.1338 - val_accuracy: 0.9564
Epoch 30/30
187/187 [=====] - 4s 32ms/step - loss: 0.0808 - accuracy: 0.9787 - val_loss: 0.1186 - val_accuracy: 0.9685
2026-04-13 00:18:18.804 | train_transformer | INFO | Saved transformer model to models/posture_transformer.keras
2026-04-13 00:18:18.804 | train_transformer | INFO | Final validation accuracy: 0.9685
```

Quantitative Results

Current Results

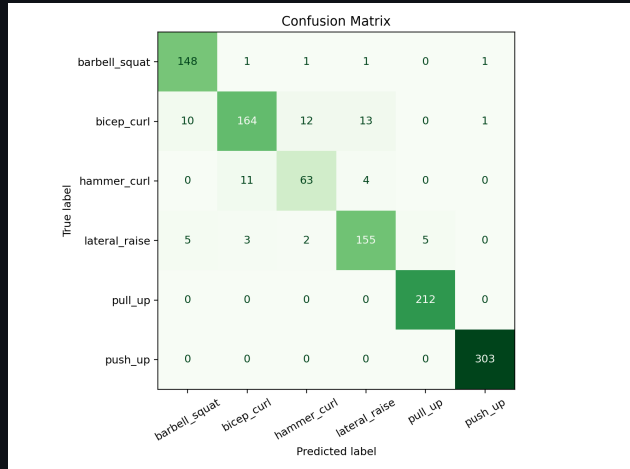
- LSTM Top-1: **84.30%**
- BiLSTM Top-1: **93.72%**
- BiLSTM Top-3: **99.82%**
- Transformer Top-1: **97.89%**

BiLSTM Quality Metrics

- Weighted F1-score: **93.64%**
- Macro F1-score: **91.69%**

Strongest classes

- push_up: F1 **0.997**
- pull_up: recall **1.00**
- barbell_squat: F1 **0.94**



How Accuracy Is Calculated

Top-1 Accuracy

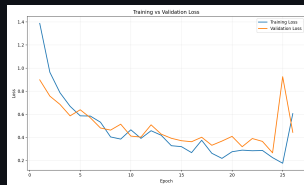
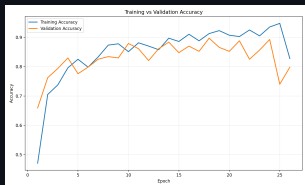
$$\text{Top-1} = \frac{\text{Correct highest-confidence predictions}}{\text{Total predictions}} \times 100$$

Top-3 Accuracy

$$\text{Top-3} = \frac{\text{True class inside top 3 predictions}}{\text{Total predictions}} \times 100$$

Precision, Recall, F1-score

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = 2 \cdot \frac{PR}{P + R}$$



Integrated Web Application

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Chapter3: Mathematical model/Architecture- System Requirements or Specification
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Implemented System Components

- FastAPI backend for live frame analysis,
- React frontend for workout and nutrition flows,
- SQL-based user progress tracking,
- meal planner, scan-to-calorie flow, and recipe CRUD,
- session reports with perfect reps, corrected reps, and common mistakes.

Improvement over surveyed papers

- not only recognition,
- not only pose correction,
- but a full usable coaching application.

Final Summary and Future Work

Key Takeaway

Future Improvements

- expand to deadlift and overhead press,
- create a stricter unseen holdout benchmark,
- improve mobile deployment,
- add multi-view support,
- improve browser-side speed and user-specific calibration.

Conclusion

- strong real-time pipeline,
- high current classification performance,
- deployable UI plus database integration,
- clear scope for improvement and scale.

Thank You

FormFix AI

Questions?